Next-generation Bayesian local robustness

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Berkeley Neyman Seminar
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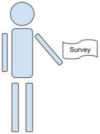
The basic problem

We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



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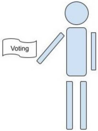
(This is the “Data”, $\mathcal{D}$)



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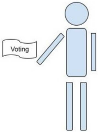

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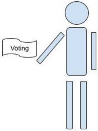

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This talk will study "Multilevel regression and poststratification" estimators [Gel97]:

- Set $\mathbb{E}\left[y|x,\theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T}\sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[y|x_j\right]$.

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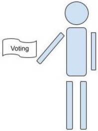

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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

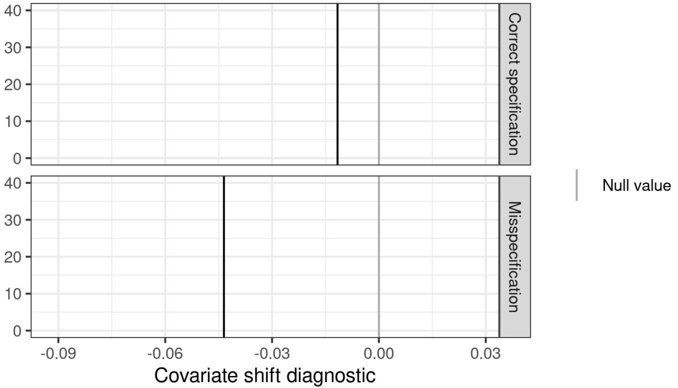

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is "large enough" to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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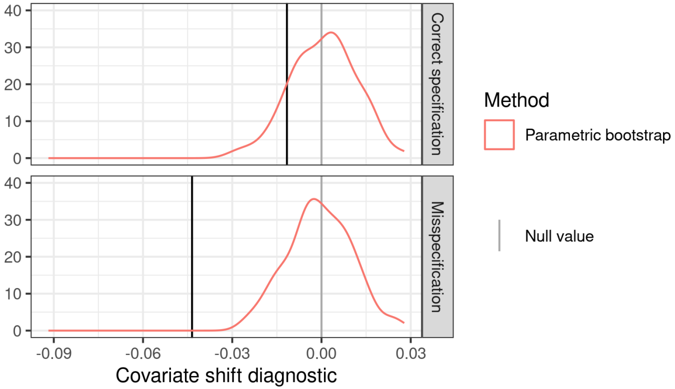

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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [GB26]

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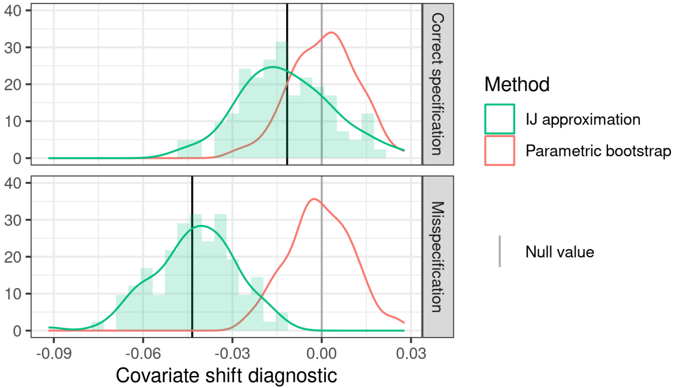

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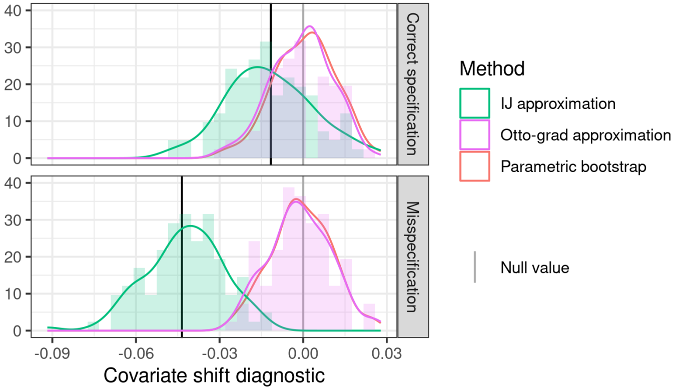

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Answer: Local glow-based sensitivity measures produce good approximations to the bootstrap at negligible computational expense.

Outline

- Local checks of specification of Bayesian regression models
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - (With Tamara Broderick)
- Bayesian robustness with local flows
  - (With Lucas Schwengber)






Local checks of specification of Bayesian regression models

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\intercal x_j) \right]}_{\text{Want to check specification without re–running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [DRM25; CBR25]
    - Bayes factors can depend strongly on priors (Lindley's paradox)
    - Articulating a family of models is inconvenient
    - Doesn't take the target into account
- Use out-of-sample checking [VGG17; Kuh+24]
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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

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$$\text{Correct specification} \quad \Rightarrow \quad \underbrace{\text{Regression estimates are invariant to } \mathcal{P}_S(x)}_{\text{This is what we actually want — let's check this directly.}}$$

Old idea in causal inference, kernel learning, transfer learning [Shi00; Büh20; Roj+; MJK20].

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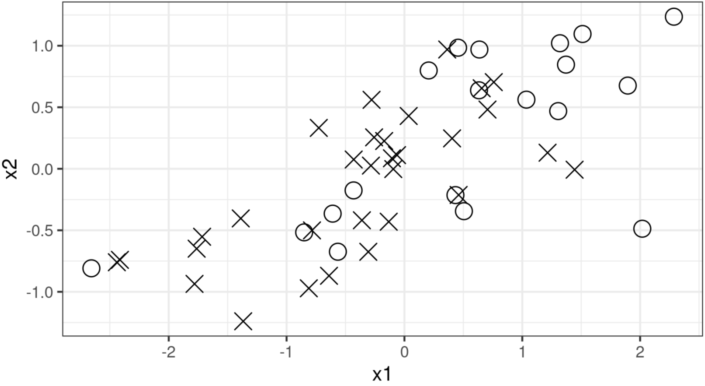
This is just a local Bayesian diagnostics version:

Let $\mathcal{P}_\mathcal{D}(x, y)$ denote a generic joint density of x, y.

1. Write our estimand as a functional $\mu(\mathcal{P}_\mathcal{D})$
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\mathcal{D}^\epsilon(x, y) = (\mathcal{P}_\mathcal{D}(x) + \epsilon w(x))\, \mathcal{P}_\mathcal{D}(y|x)$
4. Compute the derivative $\Delta(\mathcal{D}) = \partial \mu(\mathcal{P}_\mathcal{D}^\epsilon)/\partial \epsilon$ and check whether ≈ 0

Simple visual example



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Consider logistic regression with two regressors, x_1 and x_2.

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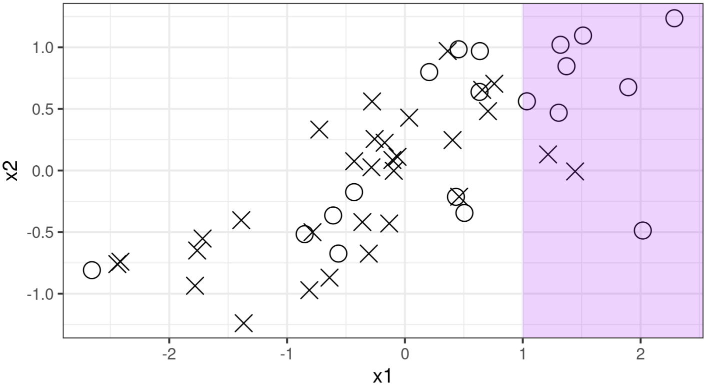

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Taking $w(x) = \mathbb{I}(x_1 > 1)$, we upweight positive examples.

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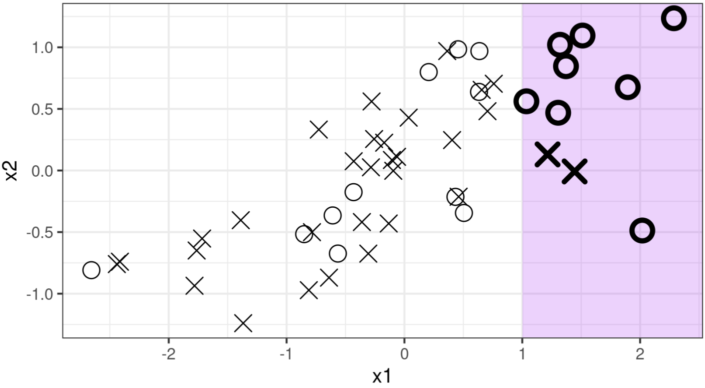

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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

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Theorem: [DF86; Gus96; GBJ18]:
Consider a family of distributions $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left.\frac{\partial \hat{\mu}}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\underbrace{\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^{\intercal} x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)}_{\text{Log model perturbation}}\right)$$

We expect $\Delta(\mathcal{D})$ to be *close* to zero when the model is correctly specified.

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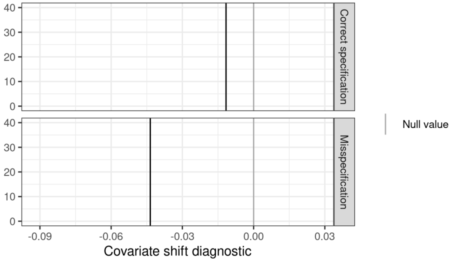

Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)}: \texttt{y} \sim \text{Logistic}(\texttt{1 + X1 + X2 + X3 + X4 + X5})$$

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But on actual data, $\Delta(\mathcal{D})$ will not be precisely zero, even under correct specification.

To assess “closeness to zero,” compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \dots, 100$:
  1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \dots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
  2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ **(time consuming!)**
  3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

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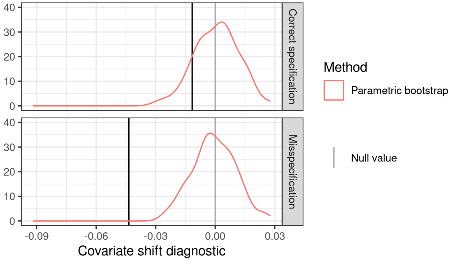

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Frequentist variance with the infinitesimal jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \overset{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

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No need to commit to our covariate shift diagnostic to care!

- Frequentist variablity of *any* model checking diagnostic is interesting
- Frequentist variablity remains meaningful under model misspecification [HM23; WDH26; LJ26; JLR25]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

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We can form a series approximation:

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\end{aligned}$$

If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von–Mises Expansion

Define the "generalized posterior" functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right) \pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right) \pi(\theta)d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

Bayesian von–Mises Expansion

Define the "generalized posterior" functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\, dy\right) \pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\, dy\right) \pi(\theta)d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, we study the *von Mises expansion* [von47].
Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t:

$$\begin{aligned}
\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) &= \sqrt{N_S}\left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S)\right) \\
&= \sqrt{N_S}\left.\frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t}\right|_{t=0}(1-0) + \mathcal{E}(\tilde{t}) \qquad (\tilde{t} \in [0,1]) \\
&= \underbrace{\sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}).
\end{aligned}$$

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem: [GB26] (sketch):
Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t}\in[0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{p} 0$.

Corollary: [von47] Let $\frac{1}{N_S}\sum_{i=1}^{N_S}(\psi_i - \overline{\psi})^2 \xrightarrow{p} V$. By the CLT applied to ψ_i,

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) \xrightarrow{d} \mathcal{N}\left(0, V\right).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S}\epsilon\left(\ell(y_i^*|x_i,\theta) - \ell(y_i|x_i,\theta)\right)\right).$$

(See, e.g., Giordano et al. [Gio+26] Theorem 4.1).

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Note: The von-Mises expansion is stronger than necessary for consistency [GB26]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

Data re-weighting.

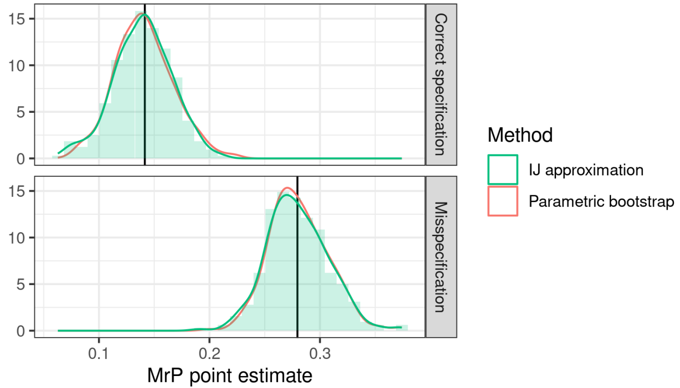

Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [LJ26] and Ji, Lee, and Rabe-Hesketh [JLR25].

Data re-weighting.

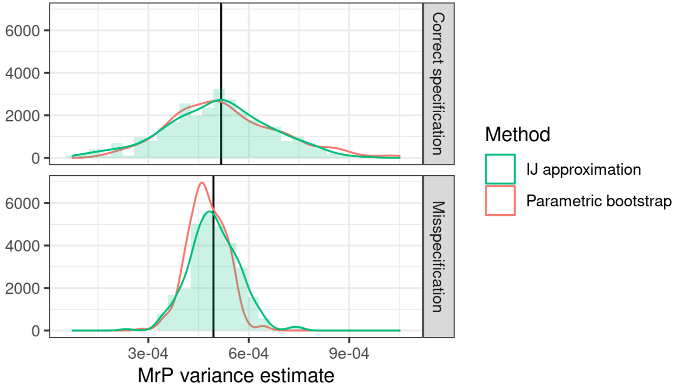

Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

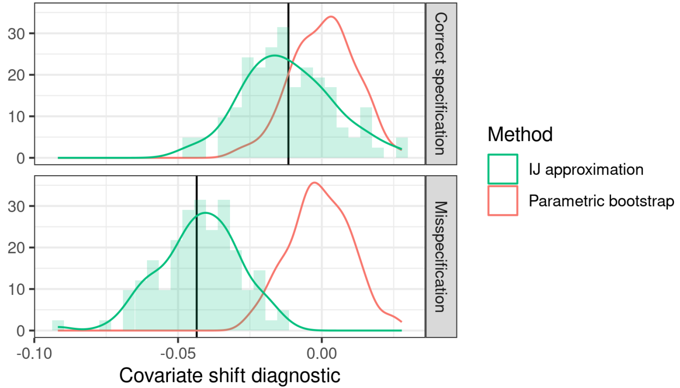

Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Bayesian robustness with local flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right] \approx \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} (\epsilon - 0),$$

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This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations [HGV25]

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Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^{\epsilon}$.

There's a large literature on learning flows between distant distributions. [Pap+21]
Our problem is easier than much of this literature:

- Only need "local" flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

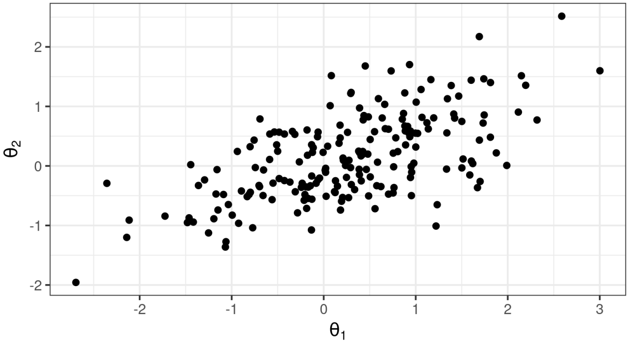

Figure 5: Draws from a posterior

Flow

Consider $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

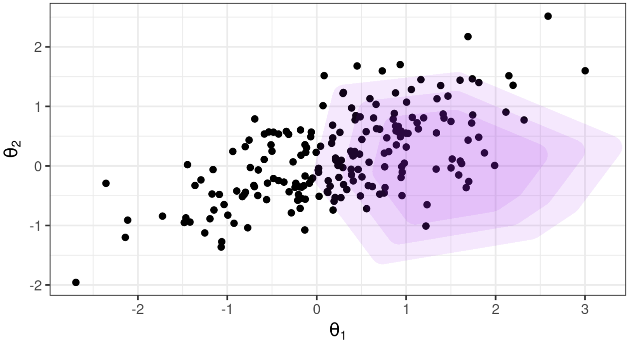

Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined "vertically."

Flow

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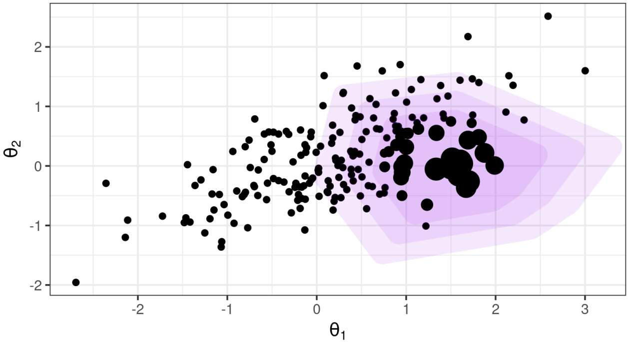

Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting [GBJ18, Appendix B]

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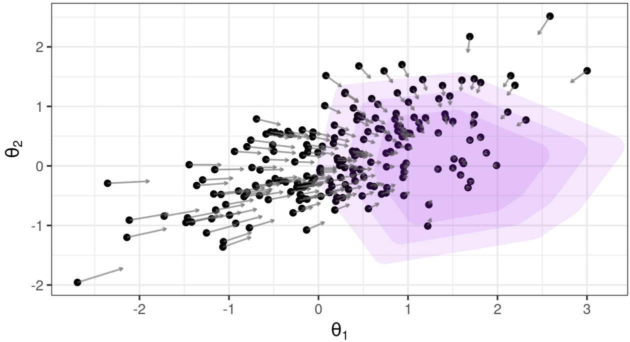

Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto G^{\epsilon}(\theta)$

Flow

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto G^\epsilon(\theta)$ such that, for small ϵ, $G^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $G^0(\theta) = \theta \sim \mathcal{P}(\theta)$.

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Suppose there is a scalar-valued potential function $\gamma(\theta)$ such that

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(\star) && \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) &= \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi(\theta)^\intercal \nabla\gamma(\theta)\right] && \text{(Connecting } h \text{ and } \gamma\text{)}
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\end{aligned}$$

Theorem: [KV86; BGL14], Schewngber 2026:
Assume that $\mathbb{E}_{\mathcal{P}(\theta)}\left[h(\theta)^2\right] < \infty$ and $\mathcal{P}(\theta)$ has sub-Gaussian tails. Then there exists a solution γ to the partial differential equation

$$\mathcal{L}_{\mathcal{P}}(\gamma) := -\nabla\gamma^\intercal\nabla\log\mathcal{P} - \Delta\gamma = h$$

such that $\|\nabla\gamma\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon\nabla\gamma(\theta)$ satisfes Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of a Langevin diffusion, and a Stein operator.[ABS+21; Son+21; Ros11]

How can we find $\gamma(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $\gamma(\theta)$ in practice?

$$(\star) \qquad \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) = \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi(\theta)^\intercal \nabla\gamma(\theta)\right] \qquad \text{(Identity connecting } h \text{ and } \gamma\text{)}$$

Basis expansion approach (the Galerkin method [EG04; Nüs24; MM24]):

- Specify a family of real-valued target functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $\gamma(\theta; \beta) := \sum_{m=1}^{M} \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

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Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi_n(\theta), h(\theta)\right) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} & \vdots & \\ \dots & \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi_n(\theta)^\intercal \nabla b_m(\theta)\right] & \dots \\ & \vdots & \end{bmatrix}}_{N \times M \text{ matrix}} = \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

An elegant flow

Theorem [GK69; BGL14], Schewngber 2026: When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $\mathcal{L}_{\mathcal{P}}^{-1}$ in its operator norm.

Using the 1^{st} order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_m$, the estimated flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\theta, h(\theta)\right)}_{\left.\frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon}\right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

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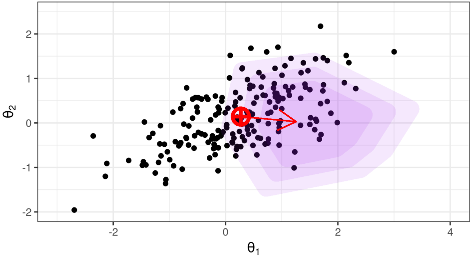

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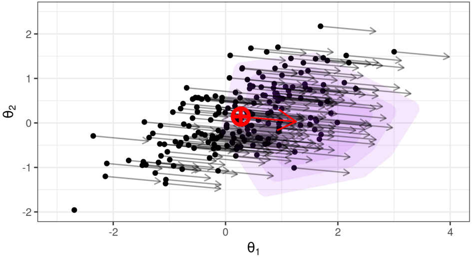

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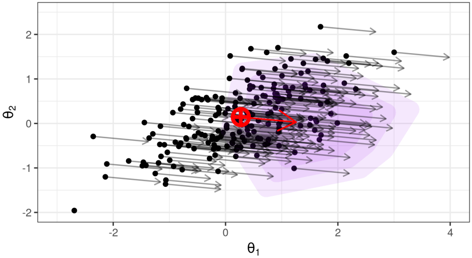

Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i.

Set $\theta_m^* = \theta_m + \frac{1}{N_S}\sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

Data re-weighting.

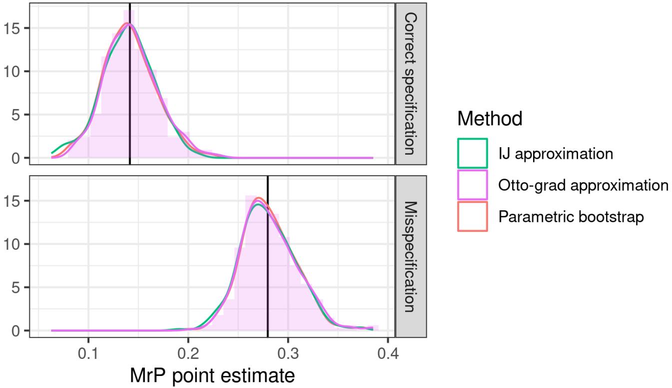

Figure 6: MrP Posterior Mean

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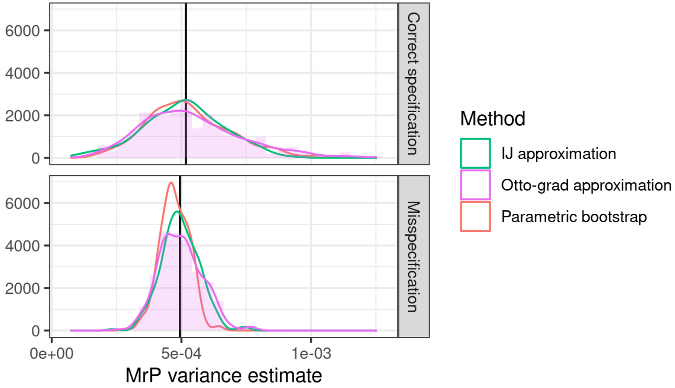

Figure 6: MrP Posterior Variance

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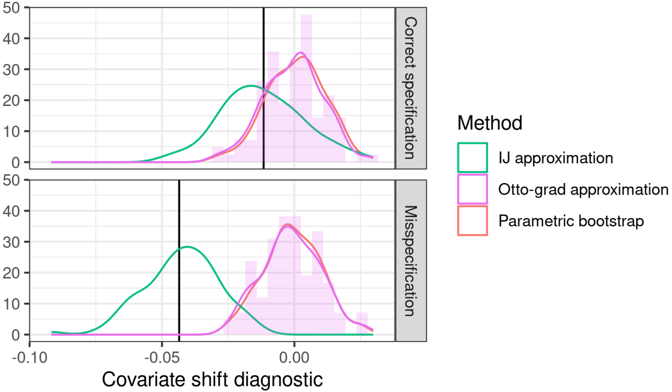

Figure 6: Covariate Shift Diagnostic

Carefully understanding the gap between IJ and the flow is ongoing work!

Conclusion

- Local checks of specification of Bayesian regression models
  - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Bayesian robustness with local flows
  - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

Lots of exciting work to to do!

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